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An Effective Deep Learning Approach for Kidney Stone Detection in CT Scan Images using Convolutional Neural Network

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***Abstract*—Kidney stones represent a significant global health concern, frequently causing individuals to seek immediate medical care due to intense pain. Radiological imaging modalities are one of the most common modes of diagnosis, but specialist interpretation is always important for accuracy. In this paper, we present computer vision in an automated kidney stone detection system that involves five stages: loading the dataset, preprocessing data, loading the InceptionV3 model, predicting, and evaluating performance. Various methods of preprocessing and data augmentation were utilized to make the model less biased and more robust. This study uses the InceptionV3 architecture to classify kidney stones in CT scan images of KUB (Kidney, Ureter, and Bladder). We utilize a total dataset of 433 participants, either diagnosed with kidney stones or in a normal state. 278 cases are diagnosed with kidney stones, and 165 are normal cases. The proposed approach achieved an impressive 98.55% accuracy across performance metrics when evaluated on an independent test dataset.**

***Index Terms*—Kidney Stone Detection, InceptionV3, Deep Learning, CT Images, CNN**

I. Introduction

The human body is a very complex mixture of all the various chemical substances needed by the human body to keep all processes such as digestion and blood flow running. Maintaining the concentration of these chemical substances within the human body is the task of two small organs known as the kidneys. The kidneys also regulate red blood cell production, rid the body of waste products through urine, and have a direct impact on blood pressure. Over the last several decades, kidney stones or renal calculi have been observed to be one of the most prevalent health disorders to plague 1 between 1% and 13% of the global population [1]. Dehydration, certain medical treatments, diet, and overweight are the leading reasons behind the occurrence of kidney stones. Different kinds of stones include those composed of uric acid, calcium oxalate, struvite, calcium phosphate etc [2]. Specific treatments are required for each kidney stone disease due to its unique characteristics [3]. Small ones pass in the urine without causing any noticeable effects to the body. When they become larger they produce symptoms including blood in the urine, nausea and painful urination. This is a picture of a kidney stone that was formed in the kidneys and urinary tract in Fig.1.

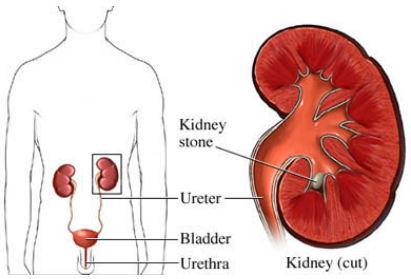

Fig. 1. Emergence of Renal Calculi in the Urinary Tract: A Pictorial Overview

Early diagnosis of kidney stones reduces the risk of complications such as urinary tract damage and infections. Any delay in treatment may lead to irrecoverable kidney damage or result in cancer [4]–[6].

In this regard, various methods have been developed and are now put into practice to disclose the undesired formation in kidneys; namely, kidney stones include computed tomography, ultrasonography, and X-ray imaging. Among them, CT scans have proved to be the most in-use modality. Nowadays, deep learning methods are able to outperform many results in different fields by taking into consideration medical images and physiological data. They had given very impressive results on lesion diagnosis [7], image segmentation of medical images [8], [9] and classification [10], [11], therefore they will be properly used for the present complicated task. Such specific applications have been varied but really excelled particularly

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in the precise delineation of kidney stones in medical images [12]. Similarly, other challenging tasks such as computational problem-solving [13], intrusion detection in IoT networks [14], [15], and disease detection including COVID-19 and lung diseases [16] have also been dealt with efficiently using DL techniques. Even hybrid meta-heuristic algorithms combined with ensemble methods like XGBoost improve intrusion detection systems [17], whereas most of the applications of DL show the broadness in the field [17].

This research investigates the efficacy of deep learning algorithms in accurately identifying kidney stones. The study, in this regard, aims at identifying kidney stones from CT scan images using deep learning models and further investigates their performances based on accuracy, precision, and recall metrics. Four experiments of performances have proven that the model InceptionNet outperformed other models in attaining an enviable 98.55% in all evaluated metrics. Data collection, image pre-processing and augmentation, deep neural network modeling, and performance evaluation are some of the major tasks performed for this research. In these experiments, 433 images of CT scans were used. Due to this small size, there may be a chance for biased results. Thus, image augmentation method was implemented in order to make the results more robust and to increase source images.

The key contributions to this research are as follows:

- An efficient automated system for detecting kidney stones has been developed. It is capable of automatically detecting kidney stones from CT scan images.
- Image augmentation was conducted to augment the original dataset images and improve the robustness of the results.

The following sections of this paper are arranged as follows: Section II discusses the literature review, the proposed methodology is detailed in Section III, and Section IV provides an overview of the results obtained from the experimental analysis. Section V discusses the state-of-the-art comparison while concluding remarks are presented in Section VI.

II. Literature Review

In the past decade, there has been widespread utilization of deep learning and machine learning methodologies for achieving precise Object Detection [18] and Image Classification. These methodologies find extensive applications in the medical domain for the expeditious diagnosis of various anomalies or illnesses [19]. The following section provides a critical overview of strategies and methods that have been adopted by previous studies associated with kidney stones' detection.

Yildirim et al. [20] presented an automated method utilizing deep learning (DL) techniques for the kidney stones detection in CT images. They performed the detection and localization of kidney stones using the XResNet-50 deep neural network. Moreover, to avoid overfitting, augmentation techniques were performed. The fine-tuning of the parameters for the XResNet-50 model was carried out through the implementation of the Adam optimizer along with cross-entropy loss. This work obtained an accuracy of 96.82% when analyzing 1799 images coming from 433 subjects. Indeed, the proposed deep model has been effective to accurately detect even diminutive kidney stones.

To address the noisy CT, Elton et al. [21] quantified the volume and identified tiny stones in coronal CT images using CNN (3D U-Net model). Subsequently, a 13-layer CNN classifier was utilized to differentiate between kidney stones and locations that were falsely identified as positive. The system had a sensitivity of 0.88 and a specificity of 0.91 for patient-level classification when evaluated on an external validation set.

Another deep neural network was presented by Irudayaraj et al. [22] that uses VGG16, ResNet50V2, MobileNetV2, and InceptionNetV3 to recognize kidney stones in CT scan images. A dataset of 1799 kidney images was used. The InceptionNet model demonstrated exceptional performance, achieving an accuracy of 86.2%, a precision of 86.6%, and a recall of 83.31%. It surpassed the other three models with improvements of 11%, 10.5%, and 2.9% in accuracy, precision, and recall, respectively.

Lopez-Tiro et al. [23] established the viability of automated kidney stones classification utilizing in-vivo endoscopic imagery. Their research evaluated six shallow machine learning techniques and three different deep learning based architectures, with the InceptionV3 model yielding the highest performance: 97% weighted accuracy, 98% recall, 97% precision, and 98% F1-Score.

Back Propagation Network (BPN) with FCM clustering was used by Akshaya et al. [24] for detection of the kidney stone, whereas Mua'ad et al. [25] carried out comparison of watershed-based, threshold-based, and edge-based segmentation techniques. Using threshold-based algorithms along with CNN, Cui et al. [26] detected kidney stones in 625 computed tomography scan images. Image segmentation is done by the authors using cascaded 3D U-Net to provide a pristine image that showed only the target region. The accuracy of the segmentation was 97% and after classification, 91% accuracy.

A research study conducted by Parakh et al. [27] used the pre-trained cascaded CNN on ImageNet and GrayNet for kidney stones detection. The authors used 535 CT images along with their segmentation, normalization, and multi-window encoding. Authors achieved 94% sensitivity and 96% specificity.

In [28], Kılıç et al. proposed a deep neural network system for kidney stone classification in DUSX images based on YOLOv4 and Mask R-CNN. To this end, they prepared a dataset of 630 images in DUSX format and defined the preprocessing procedures to be applied to enhance the clarity and detail of the images. Based on CBC preprocessing (combining BF and CLAHE), the YOLOv4 model achieves 96.1% accuracy, 99.3% precision, 96.5% recall, and an F1-score of 97.9%.

Literature shows some research gaps in the application of CNN models in medical diagnosis. First, there is underutilization of these models on diverse and larger datasets, which are very crucial to improve generalization and reduce overfitting.

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TABLE I
DATA COUNT FOR EACH CLASS

Set	Normal	Kidney Stone	Total
Training	828	625	1453
Testing	181	165	346
Total	1009	790	1799

Most of the works make use of small datasets; thus, the results have a presentation bias. Besides, training deep networks remains a challenge to low-resource institutions since large annotated datasets usually require much in terms of cost and efforts to acquire. It is also of essence that there will be an integration of multimodal data, such as the combination of CT images with patients' demographics for the development and improvement of models that will classify kidney stones with accuracy and reliability. In fact, addressing these gaps is imperative for the advancement of deep learning effectiveness in medical applications.

III. METHODOLOGY

The section is about our proposed model to predict kidney stones with high accuracy. We have proposed a transfer learning model that is based on CNN architecture: InceptionV3 to classify kidney stones in computed tomography images. We have split our methodology into a few key steps: load dataset, data pre-processing, load InceptionV3 model, prediction and evaluation of performance, shown in Fig. 2.

A. Dataset Description

In this work, the dataset was taken from [1], which consists of images taken from various sections of patients who were received at Fethi Sekin City Hospital in Turkey due to their urinary system stone disease. This dataset contained images representing various components of the urinary system, such as the kidneys, ureters, and bladder, captured through the imaging modality of CT scans. The parameters for the CT protocol included a voltage of 120 kV, an auto tube current between 100 mA and 200 mA, a 128 mm detector, a mAs value of 203, and a slice thickness of 5 mm. The images were annotated by experienced radiologists to identify the presence of urinary system stones. The dataset of CT images regarding KUB (Kidney, Ureter, and Bladder) was prepared, containing 500 patients aged from 18 to 80 years. Incorrect, certain criteria were set through which some of the cases were excluded from the dataset. A total of 67 patients were not included to the study due to the presence of double-J (pigtail) ureteral catheters, age restrictions (less than 18 or more than 80 years), having a single kidney, exhibiting kidney anomalies, or possessing atrophic kidneys. This was done to standardize the dataset so that only appropriate cases were available for analysis and further study.

B. Data Preprocessing

Data preprocessing involves identifying and addressing any errors or inaccuracies present in a dataset. In our research work, the dataset used contained CT images of kidneys acquired from different patients at different times, so unsuitable or untrustworthy data might be brought in. To make the deep learning model perform better, we need to prepare data in the most suitable way in order to increase its efficiency. The following techniques were employed to preprocess the images:

1) Images Resizing: The InceptionV3 architecture typically requires images with a minimum size of 75×75 pixels in the ImageNet Recognition Challenge, the InceptionV3 showed tremendous performance. The ImageNet database comprises over 20,000 categories of images with dimensions of 224×224 pixels that are primarily used for object recognition software. Consequently, to optimize the model's performance, the TensorFlow Keras library was utilized to resize the dataset images to 224×224 pixels.

2) Images Cropping: To identify kidney stones accurately, the region of interest needs to be located and extracted from the medical images. This basically involves the cropping of raw images to focus on the region of interest, which can greatly assist in the detection of kidney stones. Thus, by collaborating with a medical practitioner, the region of interest was identified from the raw images and highlighted for further analysis. The dimensions of the cropped were 500×285 presented in Fig. 3.

3) Data Augmentation: The data we receive in the real world may not be in the same condition as the dataset. Therefore, it is crucial to train the model with various scenarios to ensure accurate detection. To achieve this, we utilized the Imagedatagenerator class from Keras and Tensorflow for data augmentation, employing different parameters. By applying rotation (15°) angle, zooming(20%), and horizontal flip, width shift(10%), height shift(10%) to the raw images, we prevented the model from memorizing inputs and overfitting. As a result of this augmentation process, the model obtained a diverse set of angled images for classifying urinary stones shown in Fig.4.

4) Data Split: The dataset included a total of 433 individuals, with 278 individuals having stones and the remaining 165 individuals being normal. Among the individuals with stones, 218 were used for training with a total of 625 image data, while 50 were used for testing with 165 image data. As for the normal individuals, 135 were collected for training with 828 images, and 30 were used for testing with 181 images. In the testing phase, images used in the training and validation phases were excluded to prevent overfitting problems. The splitting result is shown in Table I.

C. Deep Transfer Learning Model

Transfer learning is typically a process whereby knowledge learned for a first problem can be shared to handle a related problem; this reduces the necessity for extremely large datasets and intensive training time. Deep transfer learning, in particular, employs deep neural networks to support this process, enabling the transfer of learned representations between different domains. This strategy is notably advantageous in limited labeled data, as it allows models to adapt and generalize more efficiently by leveraging existing knowledge. Moreover, Convolutional Neural Network-based deep learning classifiers

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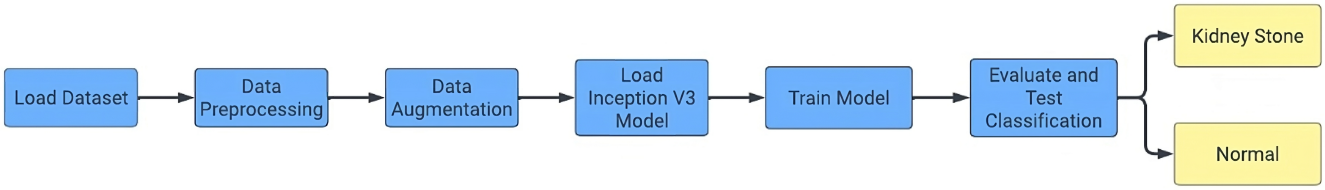

Fig. 2. The proposed methodology.

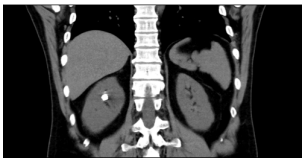

Fig. 3. Cropped CT images based on Region of Interest

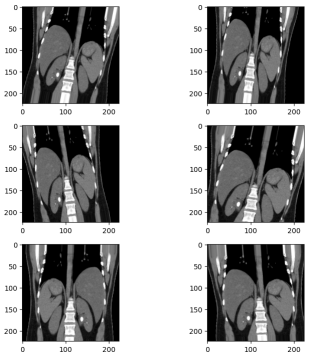

Fig. 4. CT images after Data Augmentation

have shown superior performance in tasks related to image classification and object detection [29]. In our proposed model, we utilized a deep transfer learning model, Google's Inception Convolutional Neural Network, known as InceptionV3, was first unveiled in the third edition of the ImageNet Recognition Challenge. The InceptionV3 network was trained on 1000 classes present in ImageNet, though ImageNet consists of more than 20,000 categories [30]. Inception-V3 introduces a novel approach to handling the complexity of deep learning models. This breaking down of larger convolutional layers into these smaller components results in a reduction of the overall parameter count in the model. The fewer the parameters, the lesser amount of memory and computational power one needs to work with, which helps a great deal when small data is used. Small datasets often pose a problem when training deep learning models, as these might become victims of overfitting, where the model becomes overly specialized in learning the noise rather than the patterns of the data it was trained on. Techniques such as batch normalization and dropout, being part of the design for Inception-V3, further raise its performance on smaller datasets by enforcing generalization and reducing overfitting. This overall ability to break down larger convolutional layers into smaller components makes the architecture of Inception-V3 effective in many tasks where data availability is constrained. The images of the CT scans are resized into 224×224×3 pixels and fed to the network as input for feature extraction. However, in the proposed InceoptionV3 network, the input images go through multiple convolutional layers, pooling layers and several factorization processes to extract feature maps from the images. This factorization architecture was designed to reduce the maximum number of parameters of the convolutions to make the model more optimized and cheaper than InceptionV1 [31]. In the first step, images have been allowed to pass through a 3×3 convolutional layer and produce images of size 147×147×64 pixels, followed by introducing a max pooling layer with a stride of 2. In the successive layer, convolutions are made to pass through two convolutional layers with 1×1 and 3×3 kernels along with a 3×3 max pooling operation. The generated outputs are then forwarded to the next layer with the objective of reducing parameters. In this stage, convolutions with larger spatial filters like 5×5 or 7×7 are replaced by three 1x1 and three 3×3 kernel-based convolutional layers, followed by concatenation of the outputs of convolutions filtered by 3×3 max pooling layers. This process of factorization reduces the number of parameters by 28%. The above step is repeated three times for maximum factorization. After that, results are sent to another factorization where 5×5 filters are replaced by two 3×3 filters and results are concatenated for further processing. The next-in-line layer undergoes another spatial factorization process where convolutions are divided into asymmetric convolutions. All these filters work in parallel to capture information from the images. Finally, an average pooling layer flattens the feature map to a fully connected layer, so that the model is ready for prediction. Before feeding the data into the main classifier, we utilized the ReLU activation function for the auxiliary classifier visualized in Fig.5. The presence of an auxiliary classifier in InceptionV3 promotes more stable learning and better convergence, leading to improved performance of the classifier. Since our dataset contains only two classes, we adjusted the output layer of the InceptionV3 model, which uses a softmax activation function for accurate predictions. The model underwent training for 20 epochs with batch size 15, with each epoch consisting of 46 steps. Furthermore, the InceptionV3 model's parameters were fine-tuned utilizing the Adam optimizer algorithm along with the cross-entropy loss function.

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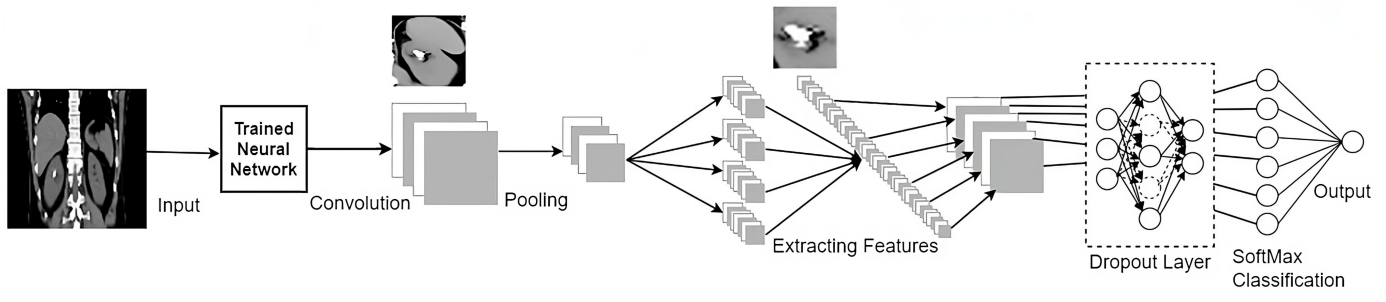

Fig. 5. InceptionV3 Architecture.

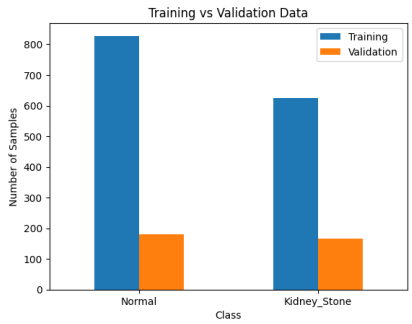

Fig. 6. Training Vs Validation Data.

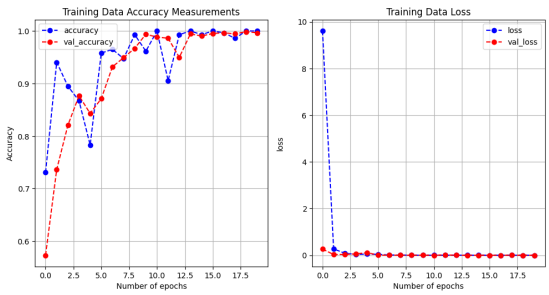

Fig. 7. Training Curves of InceptionV3 Model

IV. Experimental Result

The model was developed using the Keras library built upon the Tensorflow deep learning library in a Kaggle notebook. In order to minimize bias, data augmentation techniques were used. For training mentioned in Fig. 6, a total of 80% of the images were employed, whereas the remaining 20% were allocated for validation purposes. A graphical representation of the training process in each epoch is shown in Fig.7.

To evaluate performance, 346 unseen images were tested. The confusion matrix in Fig. 8 shows 164 kidney stone images correctly predicted as true positive and 177 images accurately classified as true negative in the testing set. However, the model mistakenly classified 4 kidney images as normal and 1 normal image as a kidney stone. Table II, shows that the model's performance was 98.55% accuracy, 97.79% precision, 99.43% recall, and 98.60% F1-score.

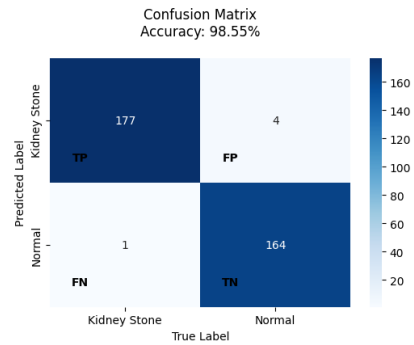

Fig. 8. Confusion Matrix Obtained on Test Data

TABLE II
Experimental Result

Method	Score(%)	
Proposed Architecture	Accuracy	98.55
	Precision	97.79
	Recall	99.43
	F1- Score	98.60

V. State of the Art Comparison

We have made an extensive comparison of our architecture with other architectures that developed earlier. The point-to-point comparison of the same is put forth in Table III. The proposed methodology yielded superior outcomes, surpassing previous studies. By incorporating region of interest extraction and data augmentation, the effectiveness of the recommended architecture saw a significant improvement.

VI. Conclusion

The study proposes a unique strategy to classifying kidney stones from KUB CT scan images using the InceptionV3 architecture. A range of data preprocessing and augmentation methods were implemented to mitigate the biasing effect, after which the model was trained. Our approach gave an overall 98.55% accuracy rate in performance metrics in an independent test dataset, which shows that our approach is effective and optimized. The high accuracy of the InceptionV3

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TABLE III
COMPARISON TABLE OF RELATED WORKS

Previous Study	Data Augmentation	Method	Accuracy (%)
Yildirim et al. [20]	No	XResNet-50	96.82
Irudayaraj et al. [22]	No	InceptionNet	86.2
Cui et al. [26]	No	CNN	97
Kılıç et al. [28]	No	YOLOv4	96.1
Lopez-Tiro et al. [23]	No	InceptionV3	97
Proposed Architecture	**Yes**	**InceptionV3**	**98.55**

model is due to its sophistication in architecture; it combines several convolutional layers with varied filter sizes, along with other modern techniques like pooling and batch normalization, that help to identify complex patterns in data and thus make the model very effective for kidney stone detection. With its reliable prediction accuracy, this model has the potential to assist radiologists in improving the kidney stones detection. Nevertheless, this research has certain limitations that should be acknowledged, particularly the lack of implementation of this technique across a variety of datasets. The findings and conclusions drawn from this study may not be fully generalizable due to this constraint, as the performance of the technique could vary significantly when applied to different types of data or in different contexts.

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